L13: Association Networks

In some cases we know the nodes of the network -- but none of the edges! Instead, we have some observations for the state of each node, and we know that the state of a node depends on the state of the nodes it is connected with.

For instance, in the context of climate science, the nodes may represent different geographical regions. For each node we may have measurements of temperature, precipitation, atmospheric pressure, etc over time. Further, we know that the climate system is interconnected, creating spatial correlations between different regions in terms of these variables (e.g., the sea surface temperate at the Indian ocean is strongly correlated with the sea surface temperature at an area of the Pacific that is west of central America). How would you construct an "association network" that shows the pairs of regions that are highly correlated in terms of each climate variable?

Recall that we had examined this problem in Lesson-8, when we discussed the δ-MAPS method. Our focus back then however was on the community detection method to identify regions with homogeneous climate variables (i.e., the nodes of the network) -- and not on the association network that interconnects those regions.

To introduce the "association network" problem more formally, suppose that we have *N* nodes, and for each node *i* we have a random vector $X_i = \begin{bmatrix} x_i^1 \\ x_i^2 \\ \vdots \\ x_i^m \end{bmatrix}$ of m *independent* observations. We want to compute an undirected network between the *N* nodes in which two nodes *i* and *j* are connected if X_i and X_j are sufficiently "associated".

To solve this problem, we need to first answer the following three questions:

a) How to measure the association between node pairs? What is an *appropriate statistical metric*?

b) Given an association metric, how can we determine whether the association between two nodes is *statistically significant*?

c) Given an answer to the previous two questions, how can we detect the set of statistically significant edges while we also *control the rate of false positives*? (i.e., spurious edges that do not really exist)

Let us start with the first question: *which association metric to use?*

The simplest association metric between $X_i = \begin{bmatrix} x_i^1 \\ x_i^2 \\ \vdots \\ x_i^m \end{bmatrix}$ and $X_j = \begin{bmatrix} x_j^1 \\ x_j^2 \\ \vdots \\ x_j^m \end{bmatrix}$ is ***Pearson's correlation coefficient***:

$\rho_{i,j} = \frac{E[(X_i-\mu_i)(X_j-\mu_j)]}{\sigma_i\, \sigma_j}$

where μ_i and σ_i are the mean and standard deviation of X_i, respectively. As you probably know, this coefficient is more appropriate for detecting linear dependencies because its absolute magnitude is equal to 1 if the two vectors are related through a (positive or negative) proportionality. If X_i and X_j are independent, then the correlation coefficient is zero (but the converse may not be true).

Instead of Pearson's correlation coefficient, we could also use *Spearman's rank correlation* (it is more robust to outliers), *mutual information* (it can detect non-linear correlations) or several other statistical association metrics. Additionally, we could use *partial correlations* to deal with the case that both X_i and X_j are affected by a third variable X_k -- the partial correlation of X_i and X_j removes the effect of X_k from both X_i and X_j.

To keep things simple, in the following we assume that the association between X_i and X_j is measured using Pearson's correlation coefficient.

Second question: ***when is the association between two nodes statistically significant?***

The "null hypothesis" that we want to evaluate is whether the correlation between X_i and X_j is zero (meaning that the two nodes should not be connected) or not:

$$H_0 : \rho_{i,j} = 0 \ \textbf{ versus } \ H_1 : \rho_{i,j} \neq 0$$

To answer this question rigorously, we need to know the statistical distribution of the metric $\rho_{i,j}$ under the null hypothesis H_0.

First let us apply the ***Fisher transformation*** on $\rho_{i,j}$ so that, instead of being limited in the range [-1,+1], it varies monotonically in $(-\infty, \infty)$:

$$z_{i,j} = \frac{1}{2} \log\left[\frac{1+\rho_{i,j}}{1-\rho_{i,j}}\right]$$

Here is a relevant result from Statistics: if the two random vectors (X_i, X_j) are uncorrelated (i.e., under the previous null hypothesis H_0) and if they follow a bivariate Gaussian distribution, the distribution of the Fisher-transformed correlation metric $z_{i,j}$ follows the Gaussian distribution with zero mean and variance $1/(m-3)$, where *m* is the dimension of the X_i vector.

Now that we know the distribution of under the null hypothesis, we can easily calculate a *p*-value for the correlation between X_i and X_j. Recall that the *p*-value represents the probability that we reject the null hypothesis H_0 even though it is true. So, we should state that X_i and X_j are significantly correlated only if the corresponding *p*-value is very small -- typically less than 1% or so.

To illustrate what we have discussed so far, the following figure shows a scatter plot of *m=445* observations for the *expression level* of two genes at the bacterium *Escherichia coli (E. coli)*. The two genes are *tyrR* and *aroG*. The expression levels were measured with microarray experiments (log-relative units). The correlation metric between the two expression level vectors is $\rho = 0.43$. Is this value statistically significant however?

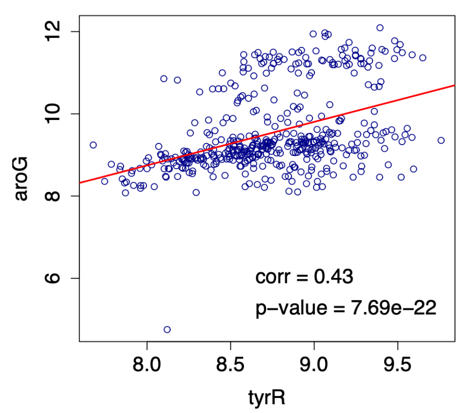

Image source: *Kolaczyk, Eric D. Statistical Analysis of Network Data: Methods and Models (2009) Springer Science+Business Media LLC.*

If we apply the Fisher transformation on ρ, we get that $z = 0.4599$. The probability that a Gaussian distribution with zero mean and variance $1/(445-3) \approx 0.0023$ gives a value of 0.4599 is less than 7.69×10^{-22}, which is also the *p*-value with which we can reject the null hypothesis H_0. So, clearly, we can infer that the two genes *tyrR* and *aroG* are significantly correlated and they should be connected in the corresponding association network.

Third question: ***how to control the rate of false positive edges?***

You may be thinking that we are done -- we now have a way to identify statistically significant correlations between node pairs. So we can connect nodes *i* and *j* with an edge if the corresponding *p*-value of the null hypothesis for $\rho_{i,j}$ is less than a given threshold (say 1%). What is the problem with this approach?

Suppose that we have $N = 1000$ nodes, and thus $m = N(N-1)/2 = 499,500$ potential edges in our network. Further, suppose that a correlation $\rho_{i,j}$ is considered significant if the *p*-value is less than 1%. Consider the extreme case that none of these pairwise correlations are actually significant. This means that if we apply the previous test 499,500 times, in 1% of those tests we will ***incorrectly*** reject the null hypothesis that $\rho_{i,j}$=0. In other words, we may end up with 4,995 spurious edges that are false positives!

This is a well-known problem in Statistics, referred to as the *Multiple Testing problem*. A common way to address it is to apply the *False Discovery Rate (FDR)* method of Benjamini and Hochberg, which aims to control the rate α of false positives. For instance, if $\alpha = 10^{-6}$, we are willing to accept only up to one-per-million false positive edges.

Specifically, suppose that we sort the *p*-values of the $m = N(N-1)/2$ hypothesis tests from lowest to highest, yielding the sequence $p_{(1)} \leq p_{(2)} \leq \ldots p_{(m)}$. The Benjamini-Hochberg method finds the highest value of $k \in \{1 \ldots m\}$ such that

$$p_{(k)} \leq \frac{k}{m}\alpha$$

if such a *p*-value exists -- otherwise $k = 0$.

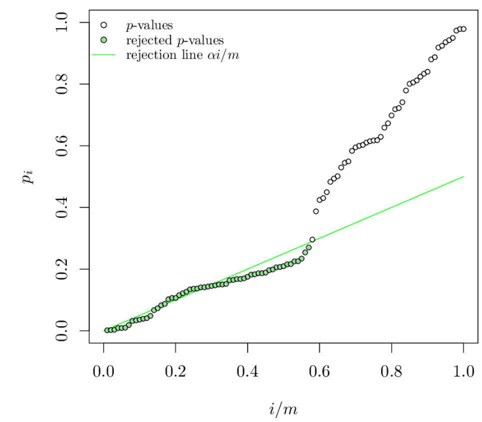

Image source: "The power of the Benjamini-Hochberg procedure", by W. van Loon, http://www.math.leidenuniv.nl/scripties/MastervanLoon.pdf

The null hypothesis for the tests with the *k* lowest *p*-values is rejected, meaning that we only "discover" those *k* edges. All other potential edges are ignored as not statistically significant. Benjamini and Hochberg proved that if the *m* tests are *independent*, then the rate of false positives in these *k* detections is less than α.

In our context, where the *m* tests are applied between all possible node pairs, the test independence assumption is typically *not* true however. There are more sophisticated FDR-control methods in the statistical literature that one could use if it is important to satisfy the α constraint.

Food For Thought

a) Explain why the m tests are probably not independent in the context of association network inference.

b) What would you do if the Benjamini-Hochberg method gives *k=0* for the value of α that you want. You are not allowed to increase α.